

JPE Replication Report

PUBLISHED

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1 Paper Identification

ID	20250778
Author	Hansen
Title	Carbon prices, forest conservation and reforestation in the Brazilian Amazon
Iteration	1

Some useful information up front:

- [JPE Data and Code Availability Policy](#)
- [Step by step guidance from Data Editor for Authors](#)
- [Template README](#)

2 Summary

In assessing compliance with our [Data and Code Availability Policy](#), our reproducibility team has identified the following issues, which we ask you to address:

1. You are required to cite all data also in the paper, not only the package readme.
2. Please add the content of `codebook.csv` to the readme - this document should be the unique source of all relevant documentation, instead of having this scattered over multiple files.
3. The description of your computational environment both local and slurm needs more detail in the readme. Locally I'm guessing you used a Mac. You need to specify exact model with specs and OS version. For the cluster versions, you need to precisely document the *exact* hardware config you asked for on the cluster when producing your results. I need to have your slurm submit scripts as part of the package, where those configs will be clearly visible. Please say exactly how many processes/cores you used, how much RAM each, and all other relevant detail.
4. Related to your usage of MacOS, we had a very hard time recreating your environment on Linux. This is of course not your fault, but I would ask you to look at our report and incorporate as many hints for future linux users as possible. We need a whole slew of system support libraries for this to work. It *did* work in the end.
5. Your readme is overall in very good shape, just needs a bit more detail.

6. All exhibits we could feasibly reproduce are fine. I need you to add a table to your readme that shows what we were able to reproduce and what not, and the reason being “computational constraints for the JPE Data Editor”. Ideally put that up front somewhere. I only need an updated version of your readme in your resubmission, leave everything else as is in the package.

3 General

- ☒ Data Availability and Provenance Statements
 - ☒ Statement about Rights - Part 1: Right to use the data (e.g. *did the authors lawfully obtain the data*)
 - ☒ Statement about Rights - Part 2: Right to publish the data (e.g. *are human subjects involved and did permission been granted to publish their data?*)
 - ☒ License for Data (optional, but recommended)
 - ☒ Details on each Data Source
- ☒ Dataset list
- ☒ Computational requirements
 - ☒ Software Requirements
 - ☐ Controlled Randomness (as necessary)
 - ☐ Memory, Runtime, and Storage Requirements
- ☒ Description of programs/code
 - ☒ License for Code (Optional, but recommended)
- ☒ List of programs
- ☐ List of tables
- ☒ References (Optional, but recommended)

4 High-level Description of Research Project

This research project...

- ☒ uses observational data.
- ☐ uses *confidential* observational data.
- ☐ uses data generated via experiment or trial by the authors (in field or lab).
 - ☐ For lab experiments: The code to implement the experimental protocol (z-Tree, o-Tree etc) is included in the package.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF document outlining the design of the experiment, including information on the selection and eligibility of participants is included.
 - ☐ A PDF copy of the instructions given to participants in both original language and an English translation is included.
- ☐ uses data generated via survey by the authors (in field or online)
 - ☐ The programs used to administer the survey (e.g. qualtrics survey file) are included.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF of the original questionnaire(s) and information on the selection and eligibility of participants is included.

- ☐ is a simulation study: the data are generated by an algorithm.
- ☐ is a numerical illustration: no data are used but code is used to produce illustrative output (tables, figures)

4.1 Data Sources

file/name	public	private	DOI/URL	Access described	cited readme	cited paper
World Bank / WDI and Climate Watch	yes	no	yes	yes	yes	no
Amazonia 2030 / Fatos da Amazonia 2021	yes	no	yes	yes	yes	no
MapBiomass Collection 5	yes	no	yes	yes	yes	no
ANA/PNRH basin layers via MapBiomass	yes	no	yes	yes	yes	no
ESA Biomass Climate Change Initiative	yes	no	yes	yes	yes	no
SEEG emissions data	yes	no	yes	yes	yes	no
IBGE agricultural census / geographies	yes	no	yes	yes	yes	yes
SEAB-PR / DERAL via	yes	no	yes	yes	yes	yes

file/name	public	private	DOI/URL	Access described	cited readme	cited paper
IPEA						
IPEA distance-to-capital data	yes	no	yes	yes	yes	no
WorldClim 2.1 climate rasters	yes	no	yes	yes	yes	yes
FGV IBRE deflator series	yes	no	yes	yes	yes	no
World Bank Carbon Pricing Dashboard	yes	no	yes	yes	yes	no

- “Data Citations” were identified from the README. However, they do not follow any specific citation format. It is recommended to make them all consistent with the JPE citation requirements with the same citation appearing in the paper and the README.
- Access conditions are described in [codebook.csv](#)

4.1.1 World Bank / WDI and Climate Watch

- ☒ Dataset is provided in public deposit
- ☐ Dataset is PRIVATELY provided (not for publication)
- ☒ DOI or URL is provided, and works
- ☒ Access conditions are described
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☐ in the manuscript
 - ☒ in the README

World Bank. World Development Indicators and World Bank source files used for Figure 1, GDP per capita, emissions, and carbon-pricing inputs. URLs: <https://data.worldbank.org/>, <https://databank.worldbank.org/source/world-development-indicators>, and <https://carbonpricingdashboard.worldbank.org/>. Access note: manuscript Figure 1 reports World Bank data were downloaded in March 2021.

World Resources Institute. Climate Watch data used with World Bank inputs for Figure 1 emissions context. URL: <https://www.climatewatchdata.org/>. Access note: retain exact downloaded extracts and metadata under replication/figure1/ and data/raw/worldbank/.

4.1.2 Amazonia 2030 / Fatos da Amazonia 2021

- ☒ Dataset is provided in public deposit
- ☐ Dataset is PRIVATELY provided (not for publication)
- ☒ DOI or URL is provided, and works
- ☒ Access conditions are described
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☐ in the manuscript
 - ☒ in the README

Amazonia 2030. 2021. Fatos da Amazonia 2021. Amazonia 2030. URL: <https://amazonia2030.org.br/>. Used for the Brazilian Amazon point in Figure

4.1.3 MapBiomass Collection 5

- ☒ Dataset is provided in public deposit
- ☐ Dataset is PRIVATELY provided (not for publication)
- ☒ DOI or URL is provided, and works
- ☒ Access conditions are described
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☐ in the manuscript
 - ☒ in the README

MapBiomass Project. Collection 5 Annual Land Cover and Land Use Maps of Brazil and related MapBiomass products. URL: <https://mapbiomas.org/>. Used for land-use/cover, total available area, agricultural area, deforestation, secondary vegetation age, pasture quality, and basin inputs. See also Souza Jr. et al. (2020), cited in the manuscript.

4.1.4 ANA/PNRH basin layers via MapBiomass

- ☒ Dataset is provided in public deposit
- ☐ Dataset is PRIVATELY provided (not for publication)
- ☒ DOI or URL is provided, and works
- ☒ Access conditions are described
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☐ in the manuscript
 - ☒ in the README

Agencia Nacional de Aguas e Saneamento Basico (ANA). 2006. National Water Resources Plan basin layers, level 2. Distributed through MapBiomass basin inputs. URL: <https://www.gov.br/ana/>. Used for basin groups in Appendix C.1.

4.1.5 ESA Biomass Climate Change Initiative

- ☒ Dataset is provided in public deposit
- ☐ Dataset is PRIVATELY provided (not for publication)
- ☒ DOI or URL is provided, and works
- ☒ Access conditions are described
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☐ in the manuscript
 - ☒ in the README

European Space Agency Climate Change Initiative Biomass Project. 2021. ESA Biomass Climate Change Initiative (Biomass_cci): Global Datasets of Forest Above-Ground Biomass for the Years 2010, 2017 and 2018, v3. NERC EDS Centre for Environmental Data Analysis. DOI: 10.5285/5f331c418e9f4935b8eb1b836f8a91b8. URL: <https://climate.esa.int/en/projects/biomass/>. Used for 2017 above-ground biomass inputs.

4.1.6 SEEG emissions data

- ☒ Dataset is provided in public deposit
- ☐ Dataset is PRIVATELY provided (not for publication)
- ☒ DOI or URL is provided, and works
- ☒ Access conditions are described
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☐ in the manuscript
 - ☒ in the README

Sistema de Estimativas de Emissões e Remoções de Gases de Efeito Estufa (SEEG). Greenhouse-gas emissions and removals data. URL: <https://seeg.eco.br/>. Used for agricultural net emissions and emissions calibration; raw filenames record a 2020.11.05 extract.

4.1.7 IBGE agricultural census / geographies

- ☒ Dataset is provided in public deposit
- ☐ Dataset is PRIVATELY provided (not for publication)
- ☒ DOI or URL is provided, and works
- ☒ Access conditions are described
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited

- ☒ in the manuscript
- ☒ in the README

4.1.8 SEAB-PR / DERAL via IPEA

- ☒ Dataset is provided in public deposit
- ☐ Dataset is PRIVATELY provided (not for publication)
- ☒ DOI or URL is provided, and works
- ☒ Access conditions are described
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☒ in the manuscript
 - ☒ in the README

Instituto de Pesquisa Economica Aplicada (IPEA). Ipeadata tables for distance-to-capital and farm-gate or commodity-price inputs. URL: <https://www.ipeadata.gov.br/>. Access notes: the manuscript reports SEAB-PR price data distributed through IPEA were accessed on February 22, 2021; the distance raw filename records an August 21, 2023 extract.

4.1.9 IPEA distance-to-capital data

- ☒ Dataset is provided in public deposit
- ☐ Dataset is PRIVATELY provided (not for publication)
- ☒ DOI or URL is provided, and works
- ☒ Access conditions are described
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☐ in the manuscript
 - ☒ in the README

Instituto de Pesquisa Economica Aplicada (IPEA). Ipeadata tables for distance-to-capital and farm-gate or commodity-price inputs. URL: <https://www.ipeadata.gov.br/>. Access notes: the manuscript reports SEAB-PR price data distributed through IPEA were accessed on February 22, 2021; the distance raw filename records an August 21, 2023 extract.

4.1.10 WorldClim 2.1 climate rasters

- ☒ Dataset is provided in public deposit
- ☐ Dataset is PRIVATELY provided (not for publication)
- ☒ DOI or URL is provided, and works
- ☒ Access conditions are described
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☒ in the manuscript
 - ☒ in the README

Fick, Stephen E., and Robert J. Hijmans. 2017. "WorldClim 2: New 1-km Spatial Resolution Climate Surfaces for Global Land Areas." *International Journal of Climatology* 37 (12): 4302–4315. DOI: 10.1002/joc.5086. Project URL: <https://www.worldclim.org/>. The package uses WorldClim 2.1, 2.5-minute historical temperature and precipitation rasters.

4.1.11 FGV IBRE deflator series

- ☒ Dataset is provided in public deposit
- ☐ Dataset is PRIVATELY provided (not for publication)
- ☒ DOI or URL is provided, and works
- ☒ Access conditions are described
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☐ in the manuscript
 - ☒ in the README

Fundacao Getulio Vargas, Instituto Brasileiro de Economia (FGV IBRE). Deflator series used for price preparation. URL: <https://portalibre.fgv.br/>. Access note: retain exact raw files and access metadata in the final archive.

4.1.12 World Bank Carbon Pricing Dashboard

- ☒ Dataset is provided in public deposit
- ☐ Dataset is PRIVATELY provided (not for publication)
- ☒ DOI or URL is provided, and works
- ☒ Access conditions are described
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☐ in the manuscript
 - ☒ in the README

World Bank. World Development Indicators and World Bank source files used for Figure 1, GDP per capita, emissions, and carbon-pricing inputs. URLs: <https://data.worldbank.org/>, <https://databank.worldbank.org/source/world-development-indicators>, and <https://carbonpricingdashboard.worldbank.org/>. Access note: manuscript Figure 1 reports World Bank data were downloaded in March 2021.

5 Requirements

- ☒ README is in TXT, MD, or PDF format
- ☒ README is accessible at the root of the package (not inside a folder)
- ☐ Title conforms to guidance (starts with "Data and Code for: Title of Paper" or "Code for: Title of Paper", is properly capitalized)

- ☐ Authors (with affiliations) are listed in the same order as on the paper

{REQUIRED} Please review the title of the deposit as per our guidelines.

{REQUIRED} Please review authors and affiliations on the deposit. In general, they are the same, and in the same order, as for the manuscript; however, authors can deviate from that order.

5.1 Stated Requirements

- ☐ No requirements specified
- ☒ Software Requirements specified as follows:
 - Python ≥ 3.9 and < 3.12
 - R with `renv`
 - Gurobi $\geq 10.0.3$
 - CmdStan through `cmdstanpy`
 - Slurm (optional)
- ☐ Computational Requirements specified as follows:
 - Cluster size, etc.
- ☒ Time Requirements specified as follows:
 - 46 hours (with Slurm parallelization)
 - 2,400 job-hours (across parallel tasks)
- ☐ Requirements are complete.

For missing requirements, see the list of required changes in [Section 8](#)

6 Analyzing Data and Code Files

6.1 Code description

- ☒ The replication package contains a "main" or "master" file(s) which calls all other auxiliary
 - programs.
 - The replication folder contains 48 R scripts, 49 Python files, and 2 shell scripts
 - The README does not identify which tables and plots are produced by what program. However, there is a separate manifest that shows the source file for the replication of each table and figure.
 - The README also does not identify the 5 in-text numbers that do not refer to tables.

NOTE: In-text numbers that reference numbers in tables do not need to be listed. Only in-text numbers that correspond to no table or figure need to be listed.

6.2 Computing Environment of the Replicator

- Nuvolos Cloud Server, "Ubuntu 24.04.1 LTS", AMD EPYC 9354P, 32 cores, 64GB RAM
- R `sessionInfo()` below:

```
R version 4.5.3 (2026-03-11)
Platform: x86_64-conda-linux-gnu
Running under: Ubuntu 24.04.1 LTS

Matrix products: default
BLAS/LAPACK: /opt/conda/lib/libopenblas-r0.3.30.so; LAPACK version 3.12.0

locale:
 [1] LC_CTYPE=en_US.UTF-8      LC_NUMERIC=C
 [3] LC_TIME=en_US.UTF-8      LC_COLLATE=en_US.UTF-8
 [5] LC_MONETARY=en_US.UTF-8  LC_MESSAGES=en_US.UTF-8
 [7] LC_PAPER=en_US.UTF-8     LC_NAME=C
 [9] LC_ADDRESS=C             LC_TELEPHONE=C
[11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C

time zone: Etc/UTC
tzcode source: system (glibc)

attached base packages:
[1] stats      graphics  grDevices  utils      datasets  methods    base

loaded via a namespace (and not attached):
[1] compiler_4.5.3
```

- R 4.5.3
- Python 3.11

7 Replication

Created a Gurobi academic account, downloaded the license, and uploaded it to the working environment.

Set the Gurobi license path in the shell:

```
export GRB_LICENSE_FILE=/files/gurobi.lic
```

7.1 Initial Environment Setup

Attempted to source the setup script in local mode:

```
source ./setup_env.sh local
```

This produced a virtual environment permission error:

```
Unable to symlink '/opt/conda/bin/python3' to '/files/package/amazon-carbon-prices/.venv/bin/python3'
bash: /files/package/amazon-carbon-prices/.venv/bin/python: Permission denied
[... additional setup output omitted ...]
```

Tried fixing the virtual environment by removing `.venv`, making the setup script executable, and rerunning setup:

```
rm -rf .venv
chmod +x ./setup_env.sh
./setup_env.sh local
```

7.2 Missing R

The setup script then reported that `Rscript` was unavailable:

```
Warning: Rscript is not available. Install R first, then rerun without --skip-renv.
```

Fixed by installing R through conda:

```
conda install -c conda-forge r-base r-essentials
```

Later, rebuilt the environment more completely with Python, R, ImageMagick, and `pkg-config`:

```
conda create -n amazon-carbon -c conda-forge python=3.11 "r-base=4.4.*" r-essentials imagemagick
conda activate amazon-carbon
```

7.3 R Package: `magick`

The `magick` package failed because the Magick++ headers/libraries were missing:

```
Error: Error installing package 'magick':
Configuration failed to find the Magick++ library.
<stdin>:1:10: fatal error: Magick++.h: No such file or directory
[... additional renv traceback omitted ...]
```

Fixed by creating a fuller conda environment with ImageMagick and `pkg-config`:

```
conda create -n amazon-carbon -c conda-forge python=3.11 "r-base=4.4.*" r-essentials imagemagick
conda activate amazon-carbon
```

7.4 R Package: `nloptr`

The `nloptr` package failed during the local NLOpt/CMake build:

```
Error: Error installing package 'nloptr':
CMake Error at CMakeLists.txt:15 (cmake_minimum_required): Compatibility with CMake < 3.5
has been removed from CMake.
/opt/conda/envs/amazon-carbon/.../ld: cannot find nlopt/lib/libnlopt.a: No such file or
directory
[... additional compiler and renv output omitted ...]
```

Fixed by installing NLOpt and build tools from conda-forge:

```
conda install -c conda-forge nlopt cmake pkg-config -y
```

7.5 R Package: `units`

The `units` package failed because `libudunits2` was missing:

```
Error: Error installing package 'units':
configure: error: Configuration failed because libudunits2.so was not found.
[... additional configure and renv output omitted ...]
```

Fixed by installing `udunits2` and `expat`:

```
conda install -c conda-forge udunits2 expat -y
```

7.6 R Package: `sf`

The first `sf` installation failed because `gdal-config` was missing:

```
Error: Error installing package 'sf':
configure: error: gdal-config not found or not executable.
[... additional renv traceback omitted ...]
```

Installed the spatial libraries:

```
conda install -c conda-forge gdal geos proj sqlite libspatialite -y
```

The next `sf` installation reached GDAL, GEOS, and PROJ, but failed compiling against the newer GDAL API:

```
Error: Error installing package 'sf':
stars.cpp:24:48: error: invalid conversion from 'CSLConstList' {aka 'const char* const*'}
to 'char**' [-fpermissive]
[... additional compiler output and renv traceback omitted ...]
```

Fixed by pinning GDAL/libGDAL to the 3.8 series:

```
conda install -c conda-forge "gdal=3.8.*" "libgdal=3.8.*" geos proj sqlite libspatialite
```

The next `sf` attempt failed because PROJ data could not be found:

Error: Error installing package 'sf':

ERROR 1: PROJ: proj_create_from_database: Open of /opt/conda/envs/amazon-carbon/share/proj failed

configure: error: OGRCoordinateTransformation() does not return a coord.trans: PROJ not available?

[... additional configure and renv output omitted ...]

Fixed by installing `proj-data` and exporting the relevant data paths:

```
conda install -c conda-forge proj proj-data -y
export PROJ_LIB=/opt/conda/envs/amazon-carbon/share/proj
export PROJ_DATA=/opt/conda/envs/amazon-carbon/share/proj
export GDAL_DATA=/opt/conda/envs/amazon-carbon/share/gdal
```

7.7 R Package: `systemfonts`

The `systemfonts` package failed because `uint32_t` was not declared:

Error: Error installing package 'systemfonts':

ft_cache.h:159:18: error: 'uint32_t' has not been declared

ft_cache.h:20:1: note: 'uint32_t' is defined in header '<cstdint>'; this is probably fixable by adding '#include <cstdint>'

[... additional compiler output and renv traceback omitted ...]

Fixed by creating an R `Makevars` file that forces inclusion of `cstdint`:

```
mkdir -p ~/.R
cat > ~/.R/Makevars <<'EOF'
CXXFLAGS += -include cstdint
CXX11FLAGS += -include cstdint
CXX14FLAGS += -include cstdint
CXX17FLAGS += -include cstdint
CXX20FLAGS += -include cstdint
EOF
```

7.8 R Package: `ragg`

The `ragg` package failed on an invalid conversion in the bundled AGG/FreeType code:

Error: Error installing package 'ragg':

agg/src/agg_font_freetype.cpp:182:35: error: invalid conversion from 'unsigned char*' to 'char*' [-fpermissive]

[... additional compiler output and renv traceback omitted ...]

Fixed by adding `-fpermissive` to the R `Makevars` C++ flags:

```
cat > ~/.R/Makevars <<'EOF'
CXXFLAGS += -include cstdint -fpermissive
CXX11FLAGS += -include cstdint -fpermissive
CXX14FLAGS += -include cstdint -fpermissive
CXX17FLAGS += -include cstdint -fpermissive
CXX20FLAGS += -include cstdint -fpermissive
EOF
```

7.9 R Package: `selectr`

The `selectr` package failed because cached `stringi` was linked against a missing ICU library:

```
Error: Error installing package 'selectr':
unable to load shared object '../stringi/libs/stringi.so': libicuuc.so.78: cannot open
shared object file: No such file or directory
[... additional lazy-loading and renv output omitted ...]
```

Fixed by removing the stale cached `stringi` package:

```
rm -rf /tmp/R/renv/cache/v5/linux-ubuntu-noble/R-4.4/x86_64-conda-linux-gnu/stringi
```

7.10 R Package: `terra`

The `terra` package initially failed because the C compiler could not create executables:

```
Error: Error installing package 'terra':
configure: error: C compiler cannot create executables
[... additional configure and renv output omitted ...]
```

Tried installing compiler/runtime packages:

```
conda install -c conda-forge compilers libstdcxx-ng libgcc-ng sysroot_linux-64 make -y
```

When the source installation failed again, installed `terra` directly from conda-forge:

```
conda install -c conda-forge r-terra -y
```

After that, `terra` installed successfully:

```
Installing terra ... OK [built from source and cached in 3.4m]
```

7.11 R Package: `tidyverse`

The `tidyverse` package failed because `stringi` was linked against a missing ICU library:

```
Error: Error installing package 'tidyverse':
unable to load shared object '.../stringi/libs/stringi.so': libicui18n.so.75: cannot open
shared object file: No such file or directory
[... additional package-loading and renv output omitted ...]
```

Fixed by removing the stale cached `stringi` package again:

```
rm -rf /tmp/R/renv/cache/v5/linux-ubuntu-noble/R-4.4/x86_64-conda-linux-gnu/stringi
```

7.12 `sf` Runtime Linkage Issue

After installing `r-terra`, `sf` failed to load because it was linked against a missing GDAL shared library:

```
Error: package or namespace load failed for 'sf':
unable to load shared object '.../sf/libs/sf.so': libgdal.so.34: cannot open shared
object file: No such file or directory
```

Fixed by removing `r-terra` and reinstalling a consistent GDAL stack:

```
conda remove r-terra -y
conda install -c conda-forge "gdal=3.8.5" "libgdal=3.8.5" geos proj proj-data sqlite libs
```

7.13 CmdStan Build

Building CmdStan failed because `TBB_CXX_TYPE` was not set for the conda compiler:

```
stan/lib/stan_math/make/libraries:114: *** "Need to set TBB_CXX_TYPE for non-standard
compiler other than gcc or clang.". Stop.
```

Fixed by setting the conda environment paths and `TBB_CXX_TYPE`, then rebuilding:

```
cd /home/datahub/.cmdstan/cmdstan-2.39.0

export CONDA_PREFIX=/opt/conda/envs/amazon-carbon
export PATH=$CONDA_PREFIX/bin:$PATH
export MAKE=$CONDA_PREFIX/bin/make
export TBB_CXX_TYPE=gcc

/opt/conda/envs/amazon-carbon/bin/make build -j1
```

- After this `setup_env.sh` ran to completion.
- The full pipeline was not run due computational limitations. A shorter run based on the README was run and finished to completion.

```
./run.sh --steps postprocess-only --backend local
```

8 Findings

8.1 Missing Requirements

- ☒ Software Requirements
 - ☒ R
 - ☒ Version 4.4
 - No packages mentioned in the README
 - ☒ Python
 - No Python packages mentioned in the README
 - ☒ GDAL
- ☒ Computational Requirements specified as follows:
 - Cluster size, disk size, memory size, etc.
 - Nothing specified in the README

[REQUIRED] Please amend README to contain complete requirements.

8.2 Data Preparation Code

Data Preparation Program	Dataset created	Reproduced?
rsrc/cleaning/_masterfile.R	data/clean/: cleaned source-specific Rdata and raster files	NA
rsrc/processing/_masterfile.R	data/processed/: cattle_price_index.Rdata; pixel_sample.Rdata; pixel_areas.Rdata; pixel_categories.Rdata; pixel_primary_forest_2017.Rdata; pixel_biomass_2017.Rdata; spatial, cross-section, and panel municipality samples; muni_biomass_2017.Rdata; muni_data.Rdata; state_emissions.Rdata; processed land-use and biome rasters	NA
rsrc/calibration/_masterfile.R	data/calibration/: calibration_1043_sites.Rdata/.csv; grid_1043_sites.geojson; calibration_78_sites.Rdata/.csv; grid_78_sites.geojson; gamma/theta calibration objects and fitted GeoJSON inputs	NA

- Already described the issues in Section 7.
- The main R code to create the processed, clean and calibration files was not run due to computation limitations.

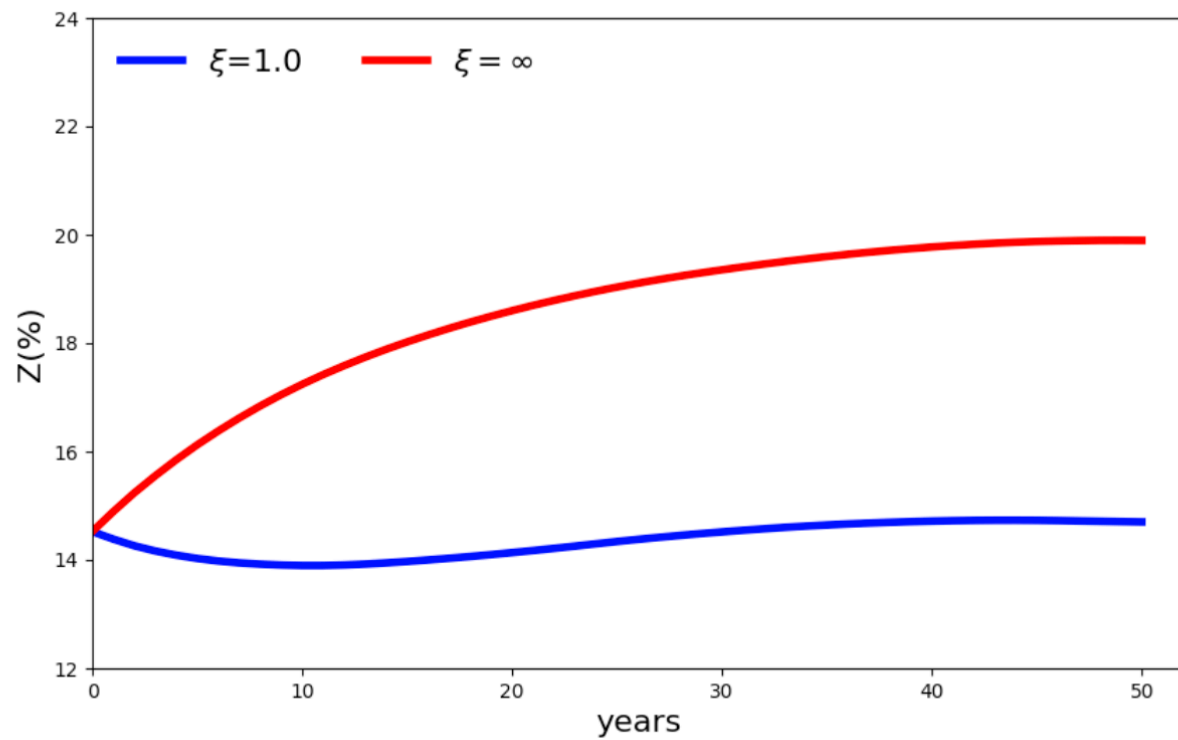
8.3 Tables and Figures

Figure/Table #	Program	Line Number	Reproduced?
Figure 1	pysrc/scripts/figure1.py	166	Yes
Figure 2	rsrc/analysis/carbon_capture_curves/02_analysis.R	68	Yes
Figure 3a	rsrc/analysis/calibration_maps_1043_sites.R	104	Yes
Figure 3b	rsrc/analysis/calibration_maps_1043_sites.R	195	Yes
Figure 4a	rsrc/analysis/calibration_maps_1043_sites.R	225	Yes
Figure 4b	rsrc/analysis/calibration_maps_1043_sites.R	246	Yes
Figure 5a	pysrc/analysis/figures.py	116	Yes
Figure 5b	pysrc/analysis/figures.py	161	Yes
Figure 6	pysrc/analysis/figures.py	557	Yes
Figure 7	rsrc/analysis/map_1043_det.R	305	Yes
Figure 8	rsrc/analysis/map_1043_det.R	353	Yes
Figure 9a	pysrc/analysis/figures.py	374	Yes
Figure 9b	pysrc/analysis/figures.py	313	Yes
Figure 9c	pysrc/analysis/figures.py	404	Yes
Figure 9d	pysrc/analysis/figures.py	344	Yes
Figure 10	pysrc/analysis/map.py	114	Yes
Figure 11	pysrc/analysis/figures.py	482	Yes
Figure 12a	pysrc/analysis/map.py	114	Yes
Figure 12b	pysrc/analysis/map.py	114	Yes
Figure 13a	pysrc/analysis/figures.py	483	Yes
Figure 13b	pysrc/analysis/figures.py	483	Yes
Figure 14a	pysrc/scripts/mpc_trajectory.py	117	Yes
Figure 14b	pysrc/scripts/mpc_trajectory.py	117	Yes

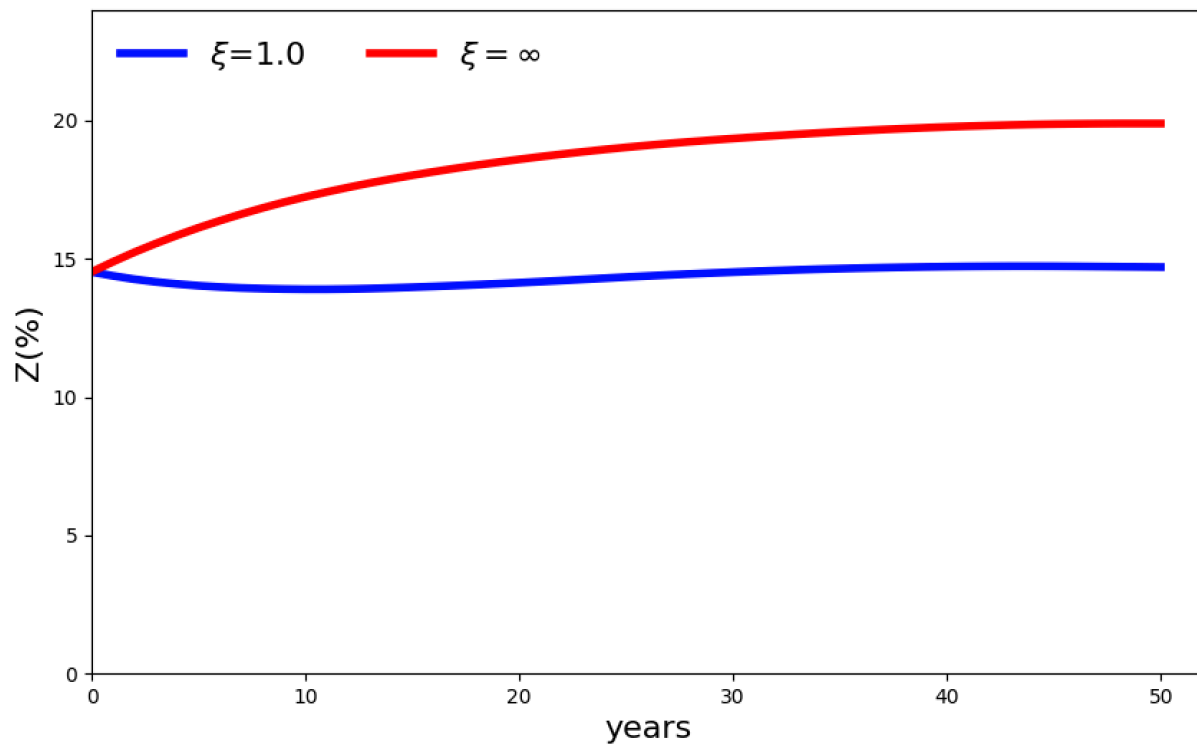
Figure/Table #	Program	Line Number	Reproduced?
Figure 15a	pysrc/scripts/price_estimation.py	130	Yes
Figure 15b	pysrc/scripts/price_estimation.py	130	Yes
Figure 16a	rsrc/analysis/map_kl.R	352	Yes
Figure 16b	rsrc/analysis/map_kl.R	366	Yes
Figure 16c	rsrc/analysis/map_kl.R	359	Yes
Figure 16d	rsrc/analysis/map_kl.R	373	Yes
Figure 17a	pysrc/analysis/figures.py	374	Yes
Figure 17b	pysrc/analysis/figures.py	313	Yes
Figure 17c	pysrc/analysis/figures.py	404	Yes
Figure 17d	pysrc/analysis/figures.py	344	Yes
Figure 18a	pysrc/analysis/figures.py	374	Yes
Figure 18b	pysrc/analysis/figures.py	313	Yes
Figure 18c	pysrc/analysis/figures.py	404	Yes
Figure 18d	pysrc/analysis/figures.py	344	Yes
Figure 19	rsrc/analysis/map_1043_hmc_xi1.R	317	Yes
Figure 20	rsrc/analysis/map_1043_hmc_xi05.R	298	Yes
Figure 21a	pysrc/scripts/bayesian_R2.py	70	Yes
Figure 21b	pysrc/scripts/bayesian_R2.py	108	Yes
Table 1	pysrc/replication/derive_carbon_prices.py	275	Yes
Table 2	pysrc/analysis/tables.py	136	Yes
Table 3	pysrc/analysis/tables.py	253	Yes
Table 4	pysrc/analysis/tables.py	454	Yes
Table 5	pysrc/replication/derive_carbon_prices.py	275	Yes
Table 6	pysrc/mpc/mpc_compute_day0.py	462	Yes

Figure/Table #	Program	Line Number	Reproduced?
Table 7	pysrc/replication/derive_mpc_transition_probabilities.py	305	Yes
Table 8	pysrc/mpc/mpc_compute_day0.py	462	Yes
Table 9	pysrc/replication/derive_mpc_transition_probabilities.py	305	Yes
Table 10	pysrc/scripts/price_estimation.py	263	Yes
Table 11	pysrc/scripts/price_estimation.py	267	Yes
Table 12	pysrc/mpc/mpc_compute_day0.py	462	Yes
Table 13	pysrc/analysis/tables.py	136	Yes
Table 14	pysrc/analysis/tables.py	258	Yes
Table 15	pysrc/analysis/tables.py	258	Yes
Table 16	pysrc/analysis/tables.py	454	Yes
Table 17	pysrc/analysis/tables.py	454	Yes
Table 18	pysrc/mpc/mpc_compute_day0.py; pysrc/mpc/mpc_compute.py	462; 178	Yes
Table 19	pysrc/mpc/mpc_compute_day0.py	462	Yes
Table 20	pysrc/replication/derive_mpc_transition_probabilities.py	305	Yes
Table 21	pysrc/mpc/mpc_compute_day0.py	462	Yes
Table 22	pysrc/replication/derive_mpc_transition_probabilities.py	305	Yes
Table 23	pysrc/sampling/baseline.py	127	Yes
Table 24	pysrc/sampling/baseline.py	116	Yes
Table 25	pysrc/sampling/baseline.py	149	Yes

- All tables and figures look similar. This is expected due to the replication only running the post-processing figure and table generation portion of the pipeline.
- Figure 11 had a minor discrepancy in scale of the y-axis. The plot and numbers match.



Paper version



Replicated version

Figure 1: Figure 11

8.4 In-Text Numbers

- ☐ In-text numbers not verified.
- ☐ There are no in-text numbers, or all in-text numbers stem from tables and figures.
- ☒ There are in-text numbers, but they are not identified in the code
- Page 17, Footnote 28: R-squared = 0.66 - could not verify location in the code
- Page 18, correlation is reported as -0.27 - same
- Page 24, Footnote 35: "our simulations show that future trajectories do not change much for b, when we move from 2 to 3 percent" - could not verify location in code
- Page 38, our calculations show that $b = 25$, $b_f = .15b$, $\tau_f = 15$, no defection would occur in 100 years - could not verify location in code
- Page 39, footnote 43 - same

9 Classification

- ☐ full reproduction
- ☐ full reproduction with minor issues
- ☒ partial reproduction (see above)
- ☐ not able to reproduce most or all of the results (reasons see above)

9.1 Reason for incomplete reproducibility

- ☐ None.
- ☐ **Discrepancy in output** (either figures or numbers in tables or text differ)
- ☐ **Bugs in code** that were fixable by the replicator (but should be fixed in the final deposit)
- ☐ **Code missing**, in particular if it prevented the replicator from completing the reproducibility check
 - ☐ **Data preparation code missing** should be checked if the code missing seems to be data preparation code
- ☐ **Code not functional** is more severe than a simple bug: it prevented the replicator from completing the reproducibility check
- ☐ **Software not available to replicator** may happen for a variety of reasons, but in particular (a) when the software is commercial, and the replicator does not have access to a licensed copy, or (b) the software is open-source, but a specific version required to conduct the reproducibility check is not available.
- ☒ **Insufficient time available to replicator** is applicable when (a) running the code would take weeks or more (b) running the code might take less time if sufficient compute resources were to be brought to bear, but no such resources can be accessed in a timely fashion (c) the replication package is very complex, and following all (manual and scripted) steps would take too long.

- ☐ **Data missing** is marked when data *should* be available, but was erroneously not provided, or is not accessible via the procedures described in the replication package
- ☐ **Data not available** is marked when data requires additional access steps, for instance purchase or application procedure.
- ☒ **Missing README** is marked if there is no README to guide the replicator, or the README is not in compliance with JPE requirements